

ELECTIVE COURSE-3.5

URA754: BRAKING AND DRIVER ASSISTANCE SYSTEMS				
	L	T	P	Cr
	3	0	0	3.0
Course Objective: This course aims to comprehensively understand vehicle dynamics, braking systems, and their components. Students will explore advanced braking technologies, including ABS, TCS, ESP, sensor technologies, and control systems, preparing them for designing and analyzing modern braking systems.				
Syllabus Vehicle Dynamics: Introduction to Vehicle dynamics, basic principles, dynamics of linear and lateral motion, Forces acting on vehicle Vehicle Braking Systems: Introduction to car-braking systems, classification, key components, brake circuit configurations, overview of braking system components Detailed Study of Braking Components: Design and function of the Brake pedal, introduction of brake servo units, master cylinder design and operation, importance of brake fluids, pipes and hoses, types of wheel brakes Advanced Braking technologies: Introduction to ABS, dynamics of braked wheel and ABS control, overview of TCS, functions and control in four-wheel drive vehicles, introduction to ESP, requirements and tasks of ESP Sensors and Control Systems in Braking: Sensor technologies, overview of automatic brake functions and its function, purpose, function and design of SBC				
Course Learning Objectives (CLO) The students will be able to: 1. Analyze the dynamics of vehicle motion and the forces influencing vehicle behavior.				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

- Understand the design, function, and classification of various vehicle braking systems and components. - Evaluate advanced braking technologies, including ABS, TCS, and ESP, and their impact on vehicle safety. - Apply knowledge of sensor technologies and control systems to enhance braking performance in modern vehicles.
Text Books Konrad Reif, Brakes, Brake Control and Driver Assistance Systems, 1st edition, Springer Vieweg Wiesbaden (2014)
Reference Books & Websites - Automotive Mechatronics: Automotive Networking, Driving Stability, Systems, Electronics, Konrad Reif, Bosch Professional Series, Springer, - Post W, Car braking systems. In: Reif, K. (eds) Fundamentals of Automotive and Engine Technology. Bosch Professional Automotive Information. Springer Vieweg, Wiesbaden (2014). - Kost, F. <i>et al.</i>, Automatic brake functions. In: Reif, K. (eds) Automotive Mechatronics. Bosch Professional Automotive Information. Springer Vieweg, Wiesbaden (2015).

Evaluation Scheme:

Sl. No	Evaluation Elements	Weightage (%)
1	MST	25
2	EST	40
3	Sessional (May include Assignments/Projects/Tutorials/Quiz/ Lab evaluations)	35

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